

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|----------------------------------|---|
| 1. Slope _____ | (A) The turning point of a V-shaped (absolute value) graph, where it changes direction. |
| 2. Zero of a function _____ | (B) A number r between -1 and 1 measuring the direction and strength of a linear association. |
| 3. Rate of change _____ | (C) An x -value where $f(x) = 0$; the graph's x -intercept. |
| 4. Vertex _____ | (D) The constant ratio $\frac{\Delta y}{\Delta x}$ — how fast y changes per unit of x . |
| 5. Axis of symmetry _____ | (E) Predicting <i>outside</i> the range of the data used to build the model. |
| 6. Piecewise function _____ | (F) A function defined by different rules on different intervals of its domain. |
| 7. Correlation coefficient _____ | (G) The steepness of a line, $\frac{\text{rise}}{\text{run}}$. |
| 8. Extrapolation _____ | (H) The vertical line $x = h$ that splits an absolute value graph into mirror halves. |

Part B — Multiple Choice (12 pts)

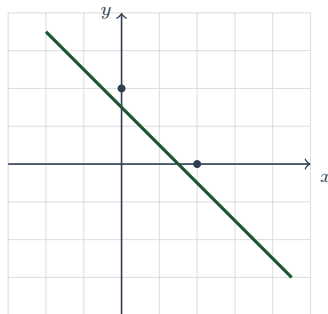
Circle the best answer.

- The slope of the line through $(-2, 1)$ and $(4, -2)$ is (A) 2 (B) $-\frac{1}{2}$ (C) -2 (D) $\frac{1}{2}$
- The graph of $y = |x - 3| + 1$ is the parent $y = |x|$ shifted (A) left 3, up 1 (B) right 3, up 1 (C) right 3, down 1 (D) left 3, down 1
- $\lfloor -2.4 \rfloor =$ (A) -2 (B) 2 (C) -3 (D) 3
- A data set has correlation coefficient $r = 0.12$. The linear association is best described as (A) strong positive (B) strong negative (C) weak/none (D) perfect
- Which equation is written in **standard form**? (A) $y = 3x - 5$ (B) $y - 2 = 3(x - 1)$ (C) $3x - y = 5$ (D) $y = |x| + 3$
- The solution of $|x| < 4$ is (A) $x < 4$ (B) $-4 < x < 4$ (C) $x < -4$ or $x > 4$ (D) $x > 4$

Part C — Short Answer & Computation (40 pts)

Show your work.

- Read the graph.** The line f is shown below.



Give the y -intercept: _____; the zero (x -intercept): _____; the slope: _____. Is f increasing or decreasing? _____

2. **Write the equation.** Find the equation of the line through $(-1, 4)$ and $(2, -2)$. Write it in slope-intercept form $y = mx + b$.

3. **Absolute value transformations.** For $g(x) = -|x - 2| + 5$, state the vertex, the axis of symmetry, whether the graph opens up or down, and the maximum or minimum value.

4. **Solve.** Solve the absolute value equation. Show both cases.

$$|2x - 5| = 9$$

5. **Solve and express three ways.** Solve $|x + 1| \leq 4$. Write the solution as an inequality and in interval notation, then graph it on the number line.

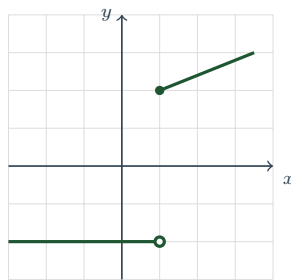


6. **Evaluate a piecewise function.** Let $f(x) = \begin{cases} 2x + 1, & x < 2 \\ 5 - x, & x \geq 2 \end{cases}$

$$f(-1) = \quad \quad \quad f(2) = \quad \quad \quad f(4) = \quad \quad \quad$$

Which piece did you use for $f(2)$, and how do you know? _____

7. **Read a piecewise graph.** The function h is shown below.



$h(-1) = \underline{\hspace{2cm}}$; $h(1) = \underline{\hspace{2cm}}$. Is h continuous at $x = 1$? Explain. $\underline{\hspace{4cm}}$

8. **Linear regression.** A study of x = hours studied and y = test score (for x between 1 and 8 hours) gives the line of best fit

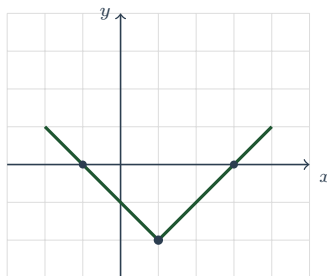
$$\hat{y} = 6.5x + 52.$$

- (a) Interpret the slope 6.5 in context. $\underline{\hspace{4cm}}$
- (b) Predict the score for a student who studies 4 hours: $\hat{y} = \underline{\hspace{2cm}}$.
- (c) Is predicting the score at $x = 20$ hours interpolation or extrapolation? Why might that prediction be unreliable? $\underline{\hspace{4cm}}$

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. The absolute value function $g(x) = |x - 1| - 2$ is graphed below.



- (a) State the two zeros of g . (b) On what interval is $g(x) < 0$ (the function negative)? On what set is $g(x) > 0$?
 (c) Using the zeros, explain *why* the sign of g switches where it does.

2. A newspaper reports that in towns with more coffee shops, residents live longer, and the correlation coefficient is $r = 0.86$. (a) Describe the direction and strength of this association. (b) Does this prove that drinking coffee makes people live longer? Explain the difference between correlation and causation, and give one lurking variable that could explain the pattern.